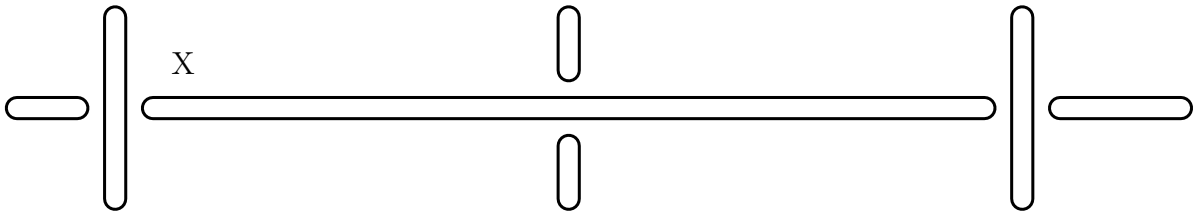
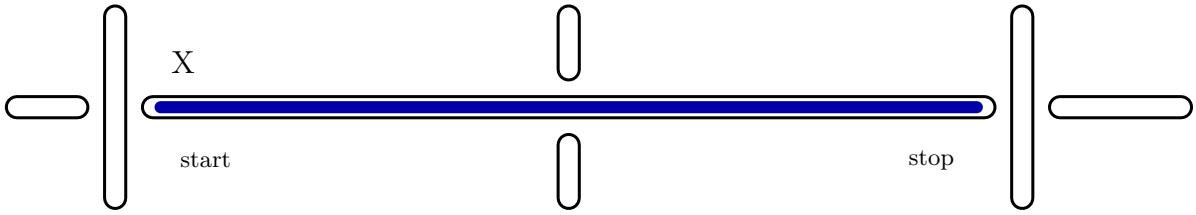
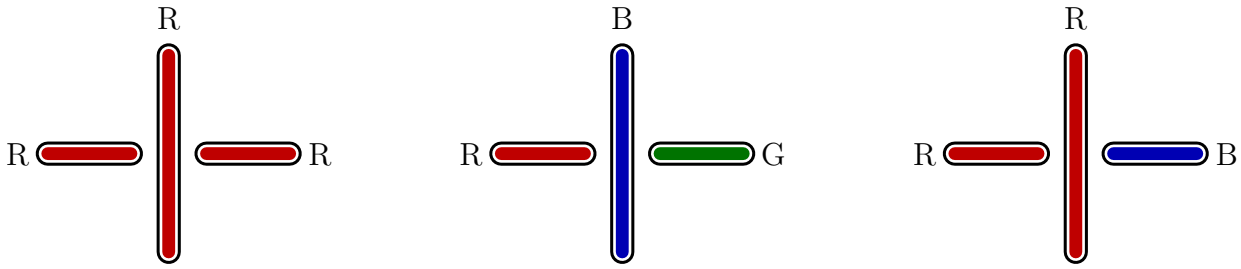
Keep cords on the table. Move them without cutting or passing strands through each other. Use red (R), blue (B), and green (G).



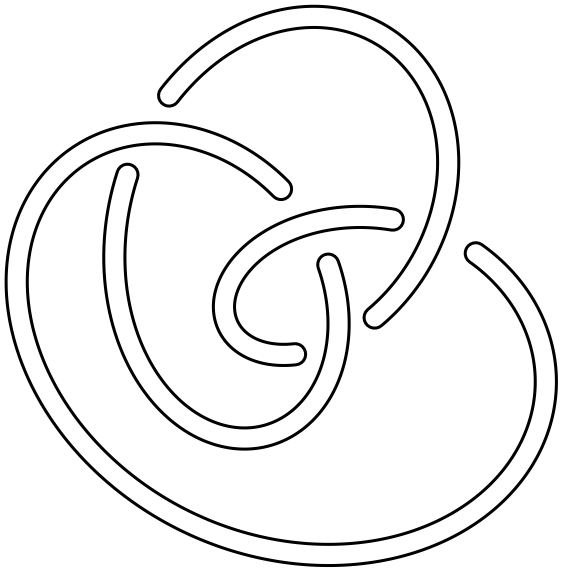

Follow X across the overpass to the next gap.




An arc runs between underpasses and continues through overpasses. A closed strand with no underpasses is one whole arc. Give each whole arc one color. At each crossing the colors must be all the same or all different.

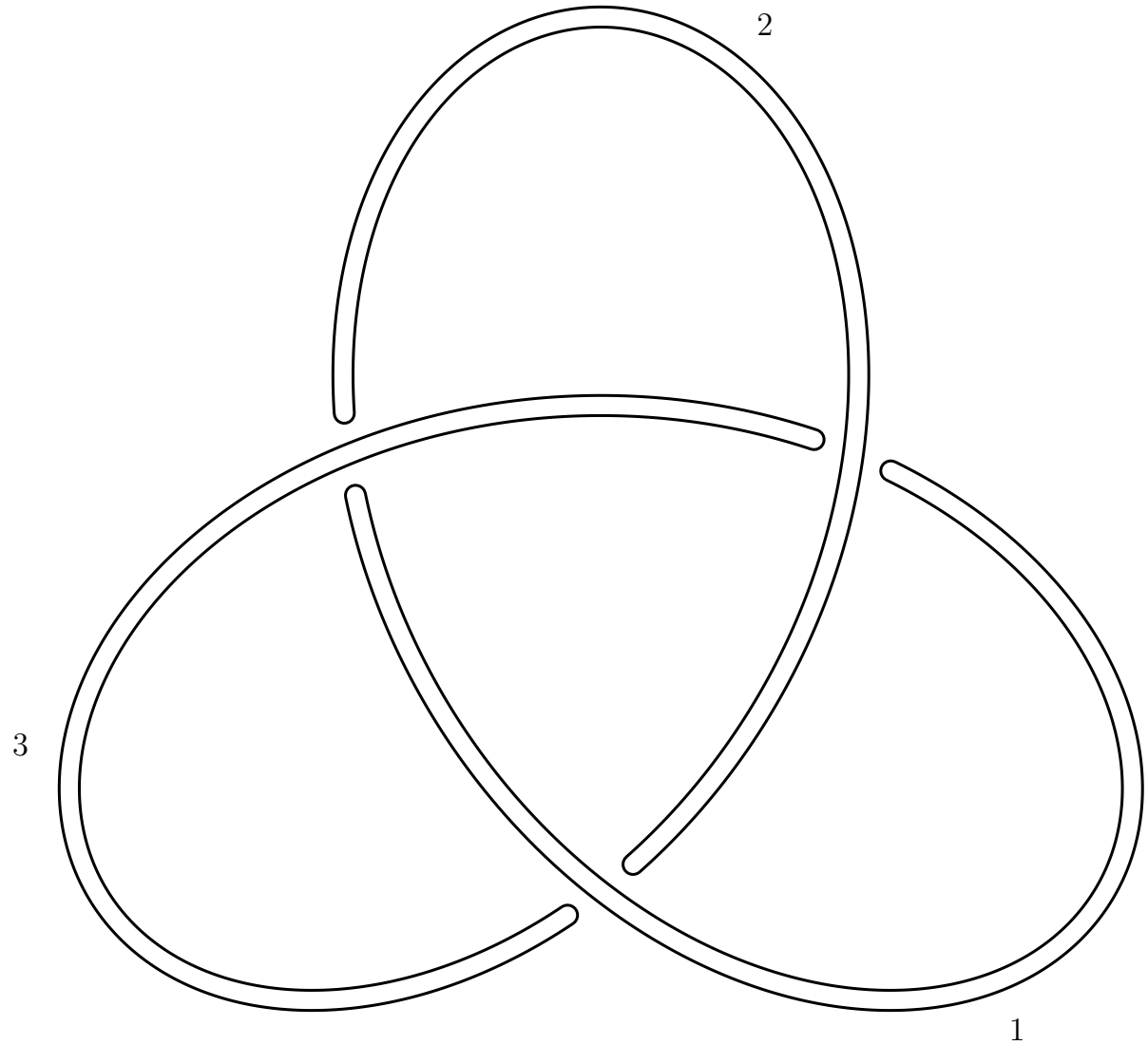


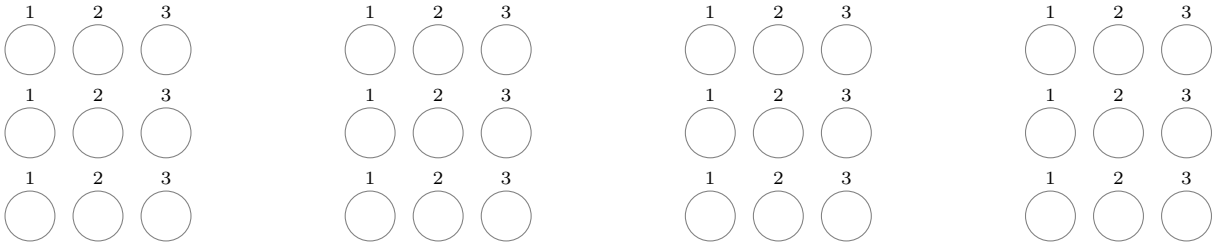

**Problem 1:** How many whole arcs does this loop have?



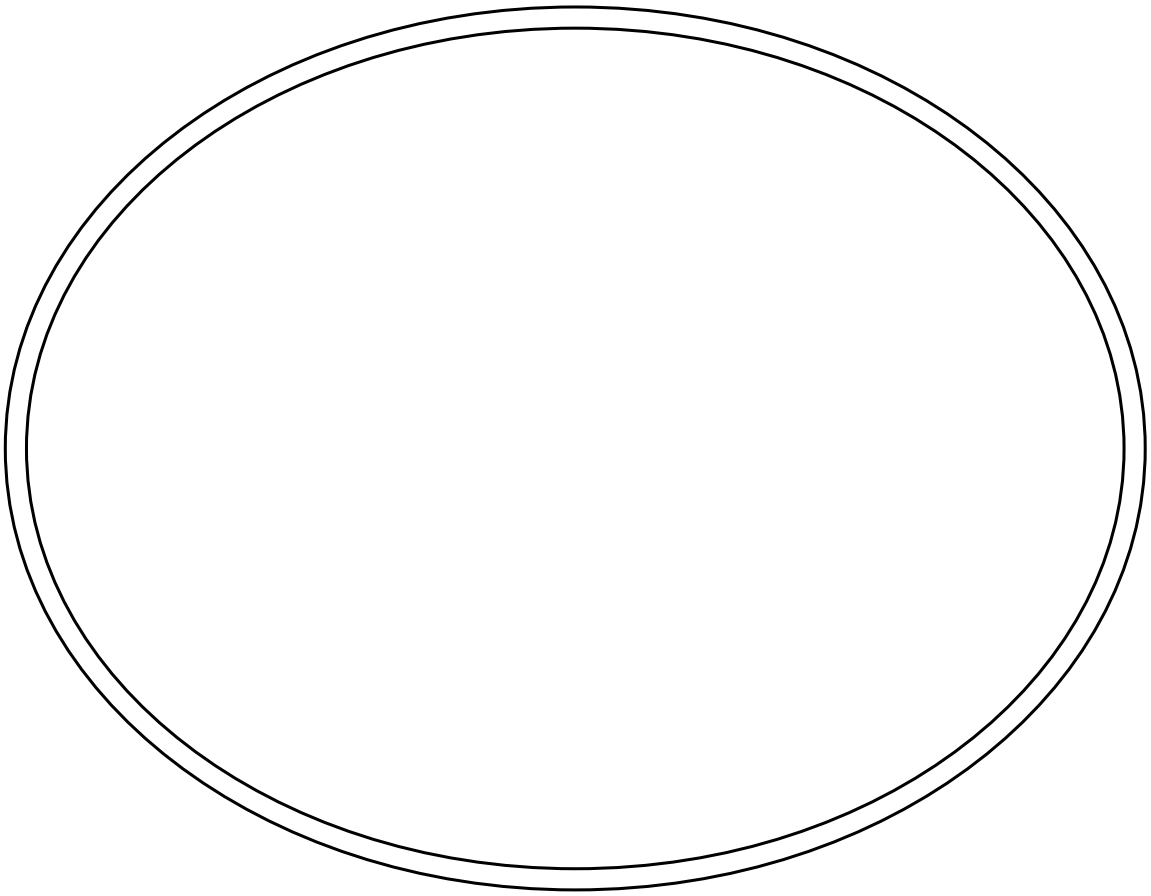
arcs: _____

**Problem 2:** Find every valid coloring of this loop, including one-color choices. Record color dots for whole arcs 1, 2, and 3; use tracing paper for new tries.



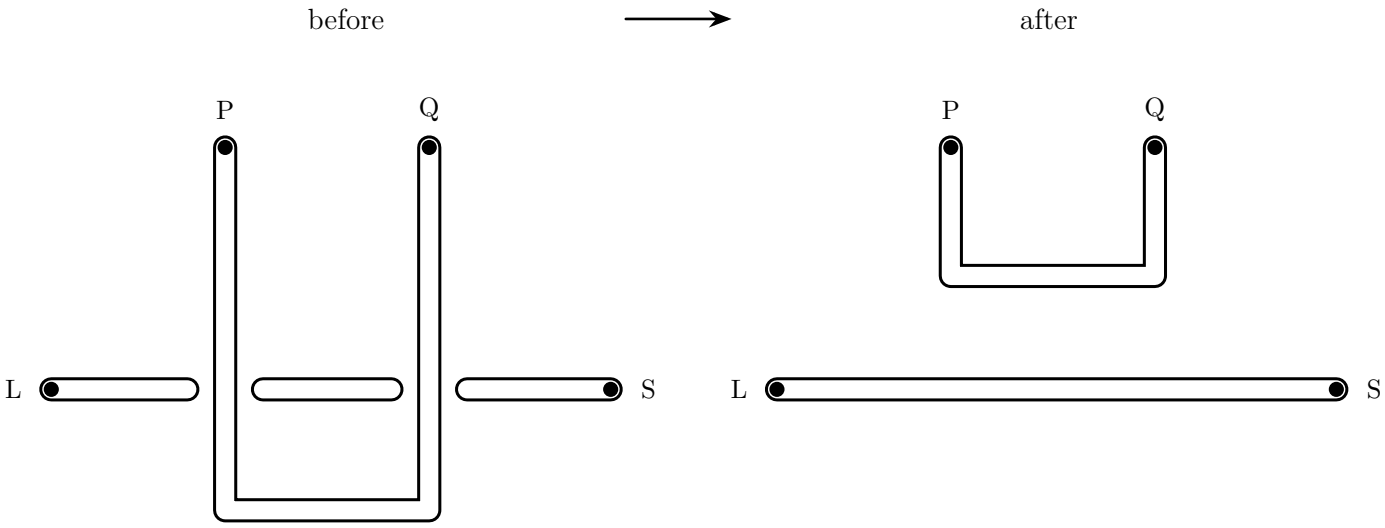



**Problem 3:** Find every valid coloring of this plain loop. Can any use more than one color?



**Problem 4:** Try each of the four closed cords A, B, C and D in the tray. Which cords can you move into this plain loop shape?

**Problem 5:** Color both pictures so matching end dots have the same colors. An arc may stop at an end dot here. Find every pair of colors at L and P that works.




| L | P | middle | S | Q |
|---|---|---|---|---|
| | | | | |
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**Problem 6:** Can the same colors at L and P give two different valid colorings of either whole picture? Explain using the crossing rule.